OFFICE OF ACADEMIC AFFAIRS

Northbridge Institute of Technology

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Academic Regulations & Student Handbook 2025–26

Department of Artificial Intelligence & Data Science
Program: M.Sc in Artificial and Big Data Analytics

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FICTIONAL TEST DOCUMENT — FOR EDU-PILOT RAG TESTING ONLY

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Program Note: Applicable Semesters III and IV correspond to the final year of this 3-year (6-semester), full-time M.Sc. program.

Table of Contents

Note: Page numbers above refer to this handbook's internal layout and are provided for navigational convenience.

1. General Academic Information

This section establishes the foundational academic framework applicable to all students enrolled in the M.Sc in Artificial and Big Data Analytics at Northbridge Institute of Technology. The regulations described herein govern program duration, the credit system, semester structure, and the conditions under which a student is considered to be in good academic standing.

1.1 Program Duration and Degree Completion

The M.Sc in Artificial and Big Data Analytics is a full-time program of **six (6) semesters** spread across **three (3) academic years**. The maximum permissible period for completion of the degree, inclusive of any authorized breaks of study, is **five (5) years** from the date of first registration. A student who does not complete all graduation requirements within this maximum period forfeits eligibility for the degree under this curriculum and must seek readmission under the curriculum in force at the time of readmission.

1.2 Credit System and Semester Load

The Institute follows a letter-graded credit system. One (1) credit corresponds to one hour of lecture (L) per week, one hour of tutorial (T) per week, or two hours of practical/laboratory (P) work per week, sustained across a fifteen-week semester. The standard semester credit load for Semester III and Semester IV is **22 credits**, distributed across theory courses, laboratory courses, and, where applicable, open electives. A student may not register for fewer than 16 credits or more than 26 credits in a single semester without written approval from the Head of Department (HoD).

1.3 Course Categories

Category	Description	Typical Credits
Core Theory	Mandatory department courses covering foundational and advanced topics in AI and Big Data Analytics.	3–4
Laboratory	Hands-on practical courses accompanying or independent of theory courses.	1–2
Program Elective	Advanced AI/Big Data Analytics courses chosen from a department-approved elective basket.	3
Open Elective	Interdisciplinary course chosen from outside the home department.	3
Mini Project / Seminar	Supervised project or literature-seminar work evaluated internally.	1–2

1.4 Registration, Withdrawal, and Academic Standing

Every student must complete semester registration, including course enrollment and fee clearance, within the registration window published in the Academic Calendar (Section 9). A registered course may be withdrawn without academic penalty only within the **add/drop period**, which closes eleven (11) calendar days after semester commencement. Withdrawal after this period requires HoD approval and results in a permanent “W” (Withdrawn) notation on the academic transcript. A student is considered to be in **Good Academic Standing** when they maintain a Cumulative Grade Point Average (CGPA) of at least 5.0 and carry no more than two active backlog courses at any time.

2. Attendance and Condonation Policy

Regular class attendance is mandatory for all students of the M.Sc in Artificial and Big Data Analytics. This section defines the minimum attendance requirement, the categories of attendance shortage, the condonation process, and the consequences of non-compliance.

2.1 Minimum Attendance Requirement

The minimum attendance requirement for examination eligibility is **75% (seventy-five percent)**. Attendance is calculated **course-wise** — that is, separately for every individually registered theory, laboratory, and elective course — rather than as a single semester-wide average. A student must independently satisfy the 75% threshold in **each** registered course to be eligible to sit the End-Semester Examination for that course, even if their overall semester attendance average is higher than 75%.

The attendance percentage for a course is computed as: *(Number of classes/practical sessions attended ÷ Number of classes/practical sessions held) × 100*, rounded to one decimal place.

2.2 Attendance Categories

Based on the course-wise attendance percentage computed under Section 2.1, every student is placed into one of the following four categories at the close of the teaching period each semester:

Attendance Range	Academic Status	Examination Eligibility	Required Action
75% and above	Eligible	Permitted to appear directly	None
65% – 74.9%	Condonable Shortage	Permitted only after condonation is granted	Submit condonation application (Section 2.3)
60% – 64.9%	Medical Condonation Only	Permitted only with approved medical condonation	Submit medical condonation with supporting certificate (Section 2.4)
Below 60%	Detained	Not permitted under any circumstance	Re-register and repeat the course in a subsequent offering

2.3 Condonation of Attendance Shortage

Students whose course-wise attendance falls in the 65%–74.9% band (“Condonable Shortage”) may apply for condonation of the shortfall subject to the following conditions:

- **Who may apply:** Any student with attendance between 65% and 74.9% in a given course, provided the shortfall is not attributable to unauthorized absence or disciplinary suspension.
- **Minimum attendance required to apply:** 65% in the relevant course.
- **Maximum shortage that can be condoned:** up to 10 percentage points below the 75% minimum (i.e., down to, but not below, 65%).
- **Approving authority:** Dean of Academic Affairs, acting on the written recommendation of the Head of Department (HoD).
- **Application deadline:** Condonation applications must be submitted no later than **5 working days** before the commencement of the End-Semester Examination for the affected course.
- **Processing fee:** A non-refundable condonation processing fee of **Rs. 500 (Rupees five hundred) per course** is payable at the time of application.

2.4 Medical Condonation

Students whose attendance falls in the 60%–64.9% band may seek **medical condonation**, which is evaluated separately from ordinary condonation under Section 2.3 and carries additional documentary requirements:

- **Acceptable medical grounds:** hospitalization, a serious illness or injury certified as incapacitating, a medically confirmed accident, or quarantine for a communicable disease as directed by a public health authority.
- **Required medical certificate:** a certificate issued on official letterhead, stating diagnosis, dates of incapacity, and fitness-to-resume date.
- **Who may issue the certificate:** a Registered Medical Practitioner (RMP) holding a valid medical council registration, or a recognized hospital's medical superintendent.
- **Hospital records requirement:** where the period of hospitalization exceeds 3 (three) consecutive days, certified hospital admission and discharge records must accompany the medical certificate.
- **Submission deadline:** the medical condonation request, together with all supporting documents, must be submitted within **7 working days** of the student resuming classes, and in any case no later than **5 working days** before the commencement of the End-Semester Examination.

Where medical condonation is approved, the effective minimum attendance requirement for examination eligibility in the affected course is relaxed from 75% to **60%**. No further relaxation below 60% is available under any circumstance.

2.5 Attendance Calculation — Additional Rules

- Attendance is recorded per scheduled lecture, tutorial, and laboratory session using the department's biometric attendance system.
- Participation in University-approved academic activities (e.g., inter-collegiate technical competitions, NSS/NCC camps, or Institute-sponsored conferences) with prior written approval from the HoD is counted as “present” for attendance purposes, subject to a cap of 10 working days per semester.
- A student who registers after the published add/drop deadline (late registration) has attendance calculated only from the effective date of registration; classes held prior to that date are excluded from both the numerator and denominator.
- For a withdrawn course (see Section 1.4), attendance calculation ceases from the date the withdrawal is formally approved, and no further sessions are counted.

2.6 Consequences of Non-Compliance

A student who remains below the applicable minimum attendance threshold after all available condonation routes (ordinary condonation under Section 2.3 and medical condonation under Section 2.4) have been exhausted, or who fails to apply within the prescribed deadlines, is formally **“Detained”** in that course.

- A Detained student is **not permitted to appear for the End-Semester Examination** in that course under any circumstance.
- The course must be re-registered and repeated in a subsequent semester in which it is offered; no grade is awarded for the detained attempt.
- Detention in three or more courses within a single semester triggers mandatory academic counselling with the HoD and may result in the student being placed on Academic Probation.
- Detention decisions may be appealed in writing to the Dean of Academic Affairs within 5 working days of the detention notice being issued, as further described in Section 10 (Attendance Grievance).

Examination eligibility rule (summary): *A student may sit the End-Semester Examination for a course only if their course-wise attendance is at least 75%, OR at least 65% with an approved ordinary condonation, OR at least 60% with an approved medical condonation. Attendance below 60% is not condonable under any process described in this handbook.*

3. Examination Regulations

This section governs the conduct of all formal assessments for the M.Sc in Artificial and Big Data Analytics, including mid-semester examinations, end-semester examinations, practical examinations, and internal assessment.

3.1 Categories of Examination

- **Mid-Semester Examination (MSE):** a 90-minute written examination conducted around the eighth week of the semester, covering approximately the first half of the syllabus, and contributing to Internal Assessment.
- **End-Semester Examination (ESE):** a 3-hour written examination conducted after completion of the teaching period, covering the full course syllabus (with emphasis on the second half), as scheduled in the Academic Calendar (Section 9).
- **Practical Examination:** a laboratory-based assessment conducted by the concerned course faculty together with an external examiner nominated by the HoD, evaluating experiments, viva-voce, and lab record maintenance.
- **Internal Assessment (IA):** continuous evaluation comprising the Mid-Semester Examination, assignments, quizzes, and class participation, as detailed in Section 4.

3.2 Examination Eligibility and Registration

Eligibility to appear for the End-Semester Examination in any course is governed by the attendance requirements of Section 2. In addition, a student must complete **online examination registration** and clear all outstanding fee dues within the window specified in the Academic Calendar. Students who fail to register within this window are treated as absent and must apply for the Supplementary Examination described in Section 3.6.

3.3 Admit Card and Identity Verification

A digital Admit Card is issued through the student portal only after attendance clearance and fee clearance are confirmed. Students must carry the Admit Card together with their Institute-issued photo ID card to every examination session. A student unable to produce valid identification will not be permitted to enter the examination hall.

3.4 Reporting Time, Late Arrival, and Absence

- Students must report to the examination hall at least 15 minutes before the scheduled start time.
- A student arriving within the **first 15 minutes** after commencement is permitted entry but is **not granted extra time**.
- A student arriving **after 15 minutes** from commencement is denied entry and is marked absent for that examination.
- No student is permitted to leave the examination hall during the first 30 minutes or the final 15 minutes of the examination.
- A student absent from the End-Semester Examination due to an emergency must notify the Central Examination Office in writing within 3 working days, supported by documentary evidence, to be considered for the Supplementary Examination.

3.5 Prohibited Devices and Examination Misconduct

The following are strictly prohibited inside the examination hall unless expressly permitted by the course-specific examination instructions: mobile phones and smartwatches, programmable calculators, Bluetooth or wireless communication devices, printed or handwritten notes not authorized as “open-book” material, and any unauthorized artificial-intelligence tool or assistant. Possession of a prohibited device, impersonation of another candidate, unauthorized collaboration, or fabrication of results constitutes examination misconduct and is addressed under Section 5 (Academic Integrity).

3.6 Medical Absence and Supplementary Examination

A student who misses the End-Semester Examination on medical grounds, supported by a certificate meeting the requirements of Section 2.4, is eligible to appear in the **Supplementary Examination**, ordinarily conducted within 6 weeks of the original End-Semester Examination. Supplementary Examination performance is graded using the same grading scale described in Section 4, with no grade-point bonus or penalty applied for having appeared supplementarily.

3.7 Backlog Examination and Revaluation

A student who fails a course (grade “F”, see Section 4.2) may clear the course through the **Backlog Examination** conducted alongside the next regular offering of the End-Semester Examination for that course, without needing to re-attend classes, **provided** the original attendance requirement under Section 2 was satisfied in the attempt that produced the fail grade. Applications for **Revaluation** of an End-Semester Examination answer script must be submitted within 10 working days of result declaration, accompanied by the prescribed revaluation fee (Section 10.7); the revalued mark, whether higher or lower, is treated as final.

4. Grading and Evaluation Policy

This section defines how course performance is converted into letter grades and how semester and cumulative performance indices are computed.

4.1 Weightage of Components

Course Type	Internal Assessment	End-Semester Examination	Practical / Continuous Evaluation
Theory Course	30%	70%	—
Laboratory Course	—	60% (End-Sem Practical)	40% (Continuous Evaluation)
Mini Project / Seminar	40%	—	60% (Evaluation Panel)

4.2 Grading Scale

The Institute uses a 10-point absolute grading scale. The minimum passing grade in every course is **"D"**, corresponding to a raw score of at least 40 marks out of 100 and a grade point of 4.

Grade	Marks Range	Grade Point	Description
O	90 – 100	10	Outstanding
A+	80 – 89	9	Excellent
A	70 – 79	8	Very Good
B+	60 – 69	7	Good
B	50 – 59	6	Above Average
C	45 – 49	5	Average
D	40 – 44	4	**Pass (minimum passing grade)**
F	Below 40	0	Fail — course must be repeated or cleared via Backlog Examination

4.3 SGPA and CGPA

The Semester Grade Point Average (SGPA) measures performance within a single semester, while the Cumulative Grade Point Average (CGPA) measures performance across all completed semesters.

SGPA Formula: SGPA = (Σ (C_i × G_i)) ÷ (Σ C_i), where C_i is the number of credits assigned to the i-th course and G_i is the grade point obtained in that course, summed across all courses registered in the semester.

CGPA Formula: CGPA = (Σ ($SGPA_k$ × TC_k)) ÷ (Σ TC_k), where $SGPA_k$ is the Semester Grade Point Average of the k-th completed semester and TC_k is the total credits registered in that semester, summed across all completed semesters up to the point of calculation.

A course in which grade “F” is obtained is included in the SGPA/CGPA computation using a grade point of 0 until the course is subsequently cleared, at which point the improved grade point replaces the “F” for all future CGPA calculations, though the original attempt remains on the transcript.

4.4 Backlog, Grade Improvement, and Revaluation

- **Backlog:** a course in which a student has received grade “F” and which remains uncleared. See Section 3.7 for the Backlog Examination process.
- **Grade Improvement:** a student who has passed a course with grade “D” or “C” may register once to reattempt the End-Semester Examination for grade improvement in a subsequent offering; the higher of the two grades is retained.
- **Revaluation:** re-checking of an End-Semester Examination answer script by a second examiner, available on payment of the prescribed fee, with the revalued mark treated as final (Section 3.7).

5. Academic Integrity and Discipline

Northbridge Institute of Technology expects the highest standards of academic honesty from every student. This section defines prohibited conduct and the associated fictional disciplinary penalties applicable within this handbook.

Category of Misconduct	Description	Penalty (First Offense)
Plagiarism	Submitting another person's work, including AI-generated text, as one's own without attribution in an assignment, project, or report.	Zero marks for the submission; formal warning recorded in academic file
Cheating in Examination	Use of unauthorized material or devices, copying from another candidate, or possession of prohibited items during an examination.	Cancellation of that examination; grade “F” in the course
Unauthorized Collaboration	Working jointly on an individually assessed assignment or examination without permission.	Zero marks for all students involved in the submission
Impersonation	Appearing for an examination or assessment on behalf of another student, or allowing another to do so.	Suspension for one full semester; grade “F” in the course
Fabrication	Falsifying laboratory data, experimental results, medical certificates, or supporting documents.	Zero marks for the affected component; disciplinary review by HoD
Assignment Copying	Substantially reproducing another student's assignment or laboratory record.	Zero marks for both the copier and the source student
Laboratory Misconduct	Unsafe handling of equipment, unauthorized use of lab resources, or falsification of lab attendance.	Suspension of laboratory access for two weeks; formal warning
Unauthorized AI Tools in Examinations	Use of AI writing, coding, or reasoning assistants during a proctored assessment where not expressly permitted.	Cancellation of that examination; grade “F” in the course

A second offense of any category listed above, or a first offense assessed as sufficiently severe by the Academic Discipline Committee, may result in suspension for up to two semesters or, in the most serious cases, recommendation for expulsion to the Institute's Disciplinary Board. Students may appeal any disciplinary finding in writing to the Dean of Academic Affairs within 10 working days of notification.

6. Semester III Curriculum

Department: Artificial Intelligence & Data Science **|** Program: M.Sc in Artificial and Big Data Analytics **|** Semester: III **|** Academic Year: 2025–26 **|** Document Type: student_handbook

6.1 ADA501 — Big Data Systems and Architectures

Course Code	ADA501	**Credits**	4
Course Type	Core Theory	**Semester**	III
Prerequisites	ADA301 — Foundations of Data Engineering	**Department**	Artificial Intelligence & Data Science

Course Objectives: Introduce distributed storage and processing architectures for large-scale data, with hands-on exposure to industry-standard big data platforms and cluster computing frameworks.

Learning Outcomes: Students will be able to design distributed data storage layouts, write distributed batch and streaming jobs, and reason about fault tolerance and scalability trade-offs.

Unit 1 — Distributed Storage Foundations

HDFS architecture, data replication, distributed file system consistency models.

Unit 2 — Batch Processing

MapReduce paradigm, Apache Spark core (RDDs, DataFrames), job scheduling.

Unit 3 — Stream Processing

Micro-batch vs. event-driven streaming, windowing, watermarking, exactly-once semantics.

Unit 4 — NoSQL and Columnar Stores

Key-value, document, and columnar data models; CAP theorem trade-offs.

Unit 5 — Cluster Resource Management

YARN/cluster schedulers, resource allocation, performance tuning.

6.2 ADA502 — Advanced Machine Learning

Course Code	ADA502	**Credits**	4
Course Type	Core Theory	**Semester**	III
Prerequisites	ADA304 — Introduction to Machine Learning	**Department**	Artificial Intelligence & Data Science

Course Objectives: Deepen understanding of supervised and unsupervised learning through ensemble methods, kernel methods, and model selection strategies applied to large, high-dimensional datasets.

Learning Outcomes: Students will be able to select, tune, and rigorously evaluate advanced machine learning models for real-world analytics problems.

Unit 1 — Model Evaluation and Selection

Bias-variance tradeoff, cross-validation, hyperparameter tuning, evaluation metrics.

Unit 2 — Ensemble Methods

Bagging, boosting (AdaBoost, gradient boosting), random forests, stacking.

Unit 3 — Kernel Methods and SVMs

Support vector machines, kernel trick, margin maximization.

Unit 4 — Unsupervised and Semi-Supervised Learning

Clustering, dimensionality reduction (PCA, t-SNE), semi-supervised techniques.

Unit 5 — Model Deployment Considerations

Feature stores, model versioning, drift detection, A/B testing basics.

6.3 ADA503 — Deep Learning

Course Code	ADA503	**Credits**	4
Course Type	Core Theory	**Semester**	III
Prerequisites	ADA304 — Introduction to Machine Learning	**Department**	Artificial Intelligence & Data Science

Course Objectives: Examine the design and training of deep neural network architectures, including convolutional and recurrent networks, and their application to structured and unstructured data.

Learning Outcomes: Students will be able to design, train, and evaluate deep neural network architectures for classification, sequence, and representation-learning tasks.

Unit 1 — Neural Network Foundations

Perceptrons, backpropagation, activation functions, loss functions.

Unit 2 — Optimization for Deep Learning

Gradient descent variants, learning rate schedules, regularization, batch normalization.

Unit 3 — Convolutional Neural Networks

Convolution/pooling operations, CNN architectures, transfer learning.

Unit 4 — Recurrent and Sequence Models

RNNs, LSTMs, GRUs, sequence-to-sequence modeling.

Unit 5 — Attention and Transformers

Self-attention mechanisms, transformer architecture, pretraining basics.

6.4 ADA504 — Statistical Foundations for Data Analytics

Course Code	ADA504	**Credits**	3
Course Type	Core Theory	**Semester**	III
Prerequisites	MA401 — Probability and Statistics	**Department**	Artificial Intelligence & Data Science

Course Objectives: Establish rigorous statistical foundations underpinning data analytics, including hypothesis testing, regression theory, and Bayesian inference.

Learning Outcomes: Students will be able to apply appropriate statistical tests and inference procedures to draw valid conclusions from analytical results.

Unit 1 — Probability Foundations

Random variables, distributions, expectation, the Central Limit Theorem.

Unit 2 — Estimation and Hypothesis Testing

Point and interval estimation, p-values, Type I/II errors, common statistical tests.

Unit 3 — Regression Theory

Linear and multiple regression, assumptions, diagnostics, regularized regression.

Unit 4 — Bayesian Inference

Bayes' theorem, priors and posteriors, Bayesian updating, credible intervals.

Unit 5 — Experimental Design

A/B testing design, sampling methods, confounding and causal inference basics.

6.5 ADA505 — Natural Language Processing

Course Code	ADA505	**Credits**	3
Course Type	Core Theory	**Semester**	III
Prerequisites	ADA304 — Introduction to Machine Learning	**Department**	Artificial Intelligence & Data Science

Course Objectives: Introduce computational techniques for processing, analyzing, and generating human language, from classical methods to modern language models.

Learning Outcomes: Students will be able to build and evaluate NLP pipelines for text classification, information extraction, and language generation tasks.

Unit 1 — Text Processing Foundations

Tokenization, stemming/lemmatization, part-of-speech tagging, n-gram models.

Unit 2 — Text Representation

Bag-of-words, TF-IDF, word embeddings (Word2Vec, GloVe).

Unit 3 — Sequence Modeling for Text

RNN/LSTM-based language models, sequence labeling, named entity recognition.

Unit 4 — Transformer-Based NLP

Attention mechanisms, pretrained language models, fine-tuning strategies.

Unit 5 — Applications

Sentiment analysis, text summarization, question answering, retrieval-augmented generation basics.

6.6 ADA506 — Cloud Computing for Data Science

Course Code	ADA506	**Credits**	3
Course Type	Program Elective	**Semester**	III
Prerequisites	ADA501 — Big Data Systems and Architectures	**Department**	Artificial Intelligence & Data Science

Course Objectives: Introduce cloud service and deployment models, managed data platforms, and distributed system design principles supporting large-scale analytics workloads.

Learning Outcomes: Students will be able to design a cloud-based data pipeline and select appropriate managed services for storage, compute, and analytics.

Unit 1 — Cloud Fundamentals

IaaS, PaaS, SaaS, public/private/hybrid cloud models.

Unit 2 — Managed Data Platforms

Managed data warehouses, data lakes, lakehouse architecture.

Unit 3 — Elastic Compute for Analytics

Auto-scaling compute clusters, serverless data processing.

Unit 4 — Cloud Data Security and Governance

Identity and access management, encryption, data cataloguing.

Unit 5 — Cost Optimization and Orchestration

Workflow orchestration tools, cost monitoring, pipeline scheduling.

6.7 ADA507 — Big Data Analytics Laboratory

Course Code	ADA507	**Credits**	1
Course Type	Laboratory	**Semester**	III
Prerequisites	Co-requisite with ADA501	**Department**	Artificial Intelligence & Data Science

Course Objectives: Provide hands-on experience implementing distributed data pipelines and analytics workloads on a cluster computing platform.

Learning Outcomes: Students will be able to build, execute, and optimize distributed data processing jobs and analyze their performance.

Unit 1 — Cluster Environment Setup Lab

Configuring a distributed processing environment and submitting basic jobs.

Unit 2 — Batch Pipeline Practice

Writing and optimizing distributed batch transformations on large datasets.

Unit 3 — Streaming Pipeline Lab

Implementing a windowed streaming aggregation pipeline.

Unit 4 — Storage Format Lab

Working with columnar storage formats and partitioning strategies.

Unit 5 — Mini Project

End-to-end big data analytics pipeline with a summary dashboard.

7. Semester IV Curriculum

Department: Artificial Intelligence & Data Science | Program: M.Sc in Artificial and Big Data Analytics | Semester: IV | Academic Year: 2025–26 | Document Type: student_handbook

7.1 ADA601 — Computer Vision

Course Code	ADA601	**Credits**	4
Course Type	Core Theory	**Semester**	IV
Prerequisites	ADA503 — Deep Learning	**Department**	Artificial Intelligence & Data Science

Course Objectives: Develop an understanding of image formation, feature extraction, and deep learning approaches to visual recognition and detection tasks.

Learning Outcomes: Students will be able to implement and evaluate deep learning models for image classification, object detection, and segmentation.

Unit 1 — Image Fundamentals

Image formation, filtering, edge detection, feature descriptors.

Unit 2 — Classical Vision Techniques

Corner detection, optical flow, classical object recognition.

Unit 3 — CNN Architectures for Vision

Landmark CNN architectures, transfer learning for vision tasks.

Unit 4 — Object Detection and Segmentation

Region-based detectors, single-shot detectors, semantic and instance segmentation.

Unit 5 — Applications

Video analytics basics, generative vision models, deployment considerations.

7.2 ADA602 — Reinforcement Learning

Course Code	ADA602	**Credits**	4
Course Type	Core Theory	**Semester**	IV
Prerequisites	ADA502 — Advanced Machine Learning	**Department**	Artificial Intelligence & Data Science

Course Objectives: Study sequential decision-making formulations and algorithms for training agents to act optimally through interaction with an environment.

Learning Outcomes: Students will be able to formulate Markov decision processes and implement value-based and policy-based reinforcement learning algorithms.

Unit 1 — MDP Foundations

Markov decision processes, rewards, policies, value functions.

Unit 2 — Dynamic Programming and Monte Carlo

Policy evaluation/improvement, Monte Carlo methods.

Unit 3 — Temporal-Difference Learning

Q-learning, SARSA, eligibility traces.

Unit 4 — Function Approximation

Deep Q-networks, policy gradient methods.

Unit 5 — Advanced Topics

Actor-critic methods, exploration strategies, multi-agent basics.

7.3 ADA603 — Data Visualization and Business Intelligence

Course Code	ADA603	**Credits**	3
Course Type	Core Theory	**Semester**	IV
Prerequisites	ADA504 — Statistical Foundations for Data Analytics	**Department**	Artificial Intelligence & Data Science

Course Objectives: Introduce principles of effective visual communication of analytical results and the design of dashboards for business decision-making.

Learning Outcomes: Students will be able to design effective visualizations and interactive dashboards that communicate analytical insight to non-technical stakeholders.

Unit 1 — Visualization Principles

Perception, visual encoding, chart selection, common pitfalls.

Unit 2 — Statistical Graphics

Distribution, comparison, and relationship visualizations.

Unit 3 — Dashboard Design

Dashboard layout, interactivity, storytelling with data.

Unit 4 — Business Intelligence Tools

BI platform architecture, data modeling for reporting, semantic layers.

Unit 5 — Communicating Analytical Results

Executive reporting, stakeholder-tailored narratives, ethical presentation of data.

7.4 ADA604 — AI Ethics and Governance

Course Code	ADA604	**Credits**	3
Course Type	Program Elective	**Semester**	IV
Prerequisites	ADA502 — Advanced Machine Learning	**Department**	Artificial Intelligence & Data Science

Course Objectives: Examine ethical, legal, and governance considerations in the design and deployment of artificial intelligence and data analytics systems.

Learning Outcomes: Students will be able to identify fairness, privacy, and accountability risks in AI systems and apply governance frameworks to mitigate them.

Unit 1 — Foundations of AI Ethics

Ethical frameworks, stakeholder impact analysis.

Unit 2 — Fairness and Bias

Sources of bias in data and models, fairness metrics, bias mitigation techniques.

Unit 3 — Privacy and Data Protection

Data minimization, anonymization, privacy-preserving analytics.

Unit 4 — Accountability and Transparency

Explainability techniques, model documentation, audit trails.

Unit 5 — Governance and Regulation

AI governance frameworks, regulatory landscape, organizational AI policy.

7.5 ADA605 — Time Series Analysis and Forecasting

Course Code	ADA605	**Credits**	3
Course Type	Program Elective	**Semester**	IV
Prerequisites	ADA504 — Statistical Foundations for Data Analytics	**Department**	Artificial Intelligence & Data Science

Course Objectives: Introduce statistical and machine learning methods for modeling and forecasting sequential, time-indexed data.

Learning Outcomes: Students will be able to build, validate, and compare forecasting models for real-world time series data.

Unit 1 — Time Series Foundations

Stationarity, decomposition, autocorrelation.

Unit 2 — Classical Forecasting Models

AR, MA, ARIMA, seasonal ARIMA.

Unit 3 — State-Space and Exponential Smoothing

Exponential smoothing methods, state-space models.

Unit 4 — Machine Learning for Forecasting

Feature-based forecasting, gradient-boosted forecasting models.

Unit 5 — Deep Learning for Sequences

RNN/LSTM-based forecasting, forecast evaluation metrics.

7.6 ADA606 — Capstone Project in AI and Big Data Analytics

Course Code	ADA606	**Credits**	2
Course Type	Mini Project / Seminar	**Semester**	IV
Prerequisites	ADA501 — Big Data Systems and Architectures	**Department**	Artificial Intelligence & Data Science

Course Objectives: Provide a capstone-style, supervised project experience integrating concepts from prior Semester III and IV courses into a working analytics system.

Learning Outcomes: Students will be able to plan, implement, document, and present a multi-week data analytics or AI project as part of a small team.

Unit 1 — Project Proposal

Problem definition, feasibility analysis, data availability assessment.

Unit 2 — Design Phase

Pipeline architecture, model selection strategy, evaluation plan.

Unit 3 — Implementation Sprint I

Data pipeline and model development with weekly supervisor review.

Unit 4 — Implementation Sprint II

Model refinement, integration, testing.

Unit 5 — Evaluation and Demonstration

Final report submission, project demonstration, and viva-voce.

8. Assignment and Laboratory Regulations

8.1 Assignment Submission

All assignments must be submitted through the department's online submission portal by the deadline specified in the course outline. Both the submission timestamp recorded by the portal and the deadline shown in the Academic Calendar (Section 9) are treated as authoritative.

8.2 Late Submission Penalty

Late assignment submissions are accepted subject to the following exact, fixed penalty: **5% of the assignment score is deducted per working day of delay, up to a maximum of five (5) working days.** Assignments submitted more than five working days after the deadline are awarded **zero marks**, except where an approved extension has been granted under Section 8.3.

8.3 Extensions

A student may request an extension of up to 3 additional working days by submitting a written request to the course instructor at least 24 hours before the original deadline. Extensions granted on medical grounds follow the documentation requirements of Section 2.4 and may extend beyond 3 working days at the instructor's discretion.

8.4 Laboratory Attendance and Records

- Laboratory attendance is calculated separately from theory attendance and is subject to the same 75% minimum threshold described in Section 2.1.
- Every student must maintain a laboratory record, to be updated and signed off by the concerned faculty at the end of each laboratory session.
- A student missing three or more consecutive laboratory sessions without prior approval is referred to the HoD for review.

8.5 Viva-Voce and Project Evaluation

Laboratory courses and mini-project courses include a mandatory viva-voce component, conducted by the course faculty together with an internal or external examiner as applicable. A student who does not appear for the scheduled viva-voce without prior approval forfeits the viva-voce marks for that evaluation cycle.

8.6 Group Assignments and Plagiarism Checking

Where an assignment is explicitly designated as a group assignment, all named group members share equal responsibility for the submitted work. All individual assignments and project reports are screened using the Institute's plagiarism detection software; a similarity index above 20% (excluding properly cited quotations and standard boilerplate) triggers review under Section 5 (Academic Integrity).

9. Academic Calendar 2025–26

9.1 Semester III — Key Dates

Milestone	Date(s)
Semester Commencement	August 4, 2025
Semester Registration Window	July 28 – August 1, 2025
Add/Drop Deadline	August 15, 2025
Mid-Semester Examination	September 22 – 26, 2025
Final Assignment Submission Deadline	November 7, 2025
Practical Examination	November 17 – 21, 2025
End-Semester Examination	November 24 – December 5, 2025
Result Declaration	December 22, 2025

9.2 Semester IV — Key Dates

Milestone	Date(s)
Semester Commencement	January 5, 2026
Semester Registration Window	December 29, 2025 – January 2, 2026
Mid-Semester Examination	February 23 – 27, 2026
Project Evaluation (ADA606)	April 13 – 17, 2026
Practical Examination	April 20 – 24, 2026
End-Semester Examination	April 27 – May 8, 2026
Result Declaration	May 26, 2026

All dates above are fictional and specific to this test handbook; they are internally consistent across Semester III and Semester IV and should not be interpreted as referring to any real institution's calendar.

10. Student Services and Administrative Procedures

10.1 Bonafide Certificate

Responsible Office	AI & Data Science Academic Office
Required Documents	Filled request form; valid ID card
Submission Process	Submit request form at the AI & Data Science Academic Office counter or via the student portal.
Processing Time	3 working days
Fee	No fee

10.2 Transcript Request

Responsible Office	Central Examination Office
Required Documents	Filled request form; ID card; fee receipt
Submission Process	Apply online through the student portal and pay the applicable fee.
Processing Time	7 working days
Fee	Rs. 250 per copy

10.3 ID Card Replacement

Responsible Office	Student Academic Services
Required Documents	Police/loss report (if lost); ID replacement form
Submission Process	Submit form along with supporting documents at Student Academic Services.
Processing Time	5 working days
Fee	Rs. 200

10.4 Course Withdrawal

Responsible Office	AI & Data Science Academic Office
Required Documents	Withdrawal form; HoD approval
Submission Process	Submit withdrawal form before the deadline in Section 1.4; obtain HoD signature.
Processing Time	2 working days for processing
Fee	No fee within add/drop period

10.5 Semester Registration

Responsible Office	AI & Data Science Academic Office

Required Documents	Fee clearance receipt; course registration form
Submission Process	Complete registration through the student portal within the window in Section 9.
Processing Time	Immediate (online)
Fee	As per fee schedule

10.6 Examination Form Submission

Responsible Office	Central Examination Office
Required Documents	Attendance clearance; fee receipt
Submission Process	Submit the examination registration form via the student portal.
Processing Time	Immediate (online)
Fee	Rs. 100 per course (late fee Rs. 300)

10.7 Grade Revaluation

Responsible Office	Central Examination Office
Required Documents	Revaluation request form; result copy
Submission Process	Submit within 10 working days of result declaration (Section 3.7).
Processing Time	15 working days
Fee	Rs. 600 per answer script

10.8 Attendance Grievance

Responsible Office	Dean of Academic Affairs
Required Documents	Grievance form; supporting evidence
Submission Process	Submit written appeal within 5 working days of the detention notice (Section 2.6).
Processing Time	10 working days
Fee	No fee

10.9 Academic Grievance

Responsible Office	Dean of Academic Affairs
Required Documents	Grievance form; supporting documentation
Submission Process	Submit written grievance describing the academic concern with evidence.
Processing Time	10 working days
Fee	No fee

11. Important Contacts

Office	Contact Person	Room	Working Hours	Email (Fictional)
AI & Data Science Academic Office	Ms. Ananya Kulkarni, Academic Coordinator	AI & Data Science Block, Room 104	9:30 AM – 5:00 PM (Mon–Fri)	cse.academic@nit-fictional.edu
Central Examination Office	Mr. Rohan Deshpande, Deputy Controller of Examinations	Admin Block, Room 210	9:00 AM – 5:30 PM (Mon–Sat)	exams@nit-fictional.edu
Student Academic Services	Ms. Priya Menon, Student Services Officer	Admin Block, Room 118	9:30 AM – 5:00 PM (Mon–Fri)	studentservices@nit-fictional.edu
Office of the Dean, Academic Affairs	Dr. Vikram Sharma, Dean of Academic Affairs	Admin Block, Room 301	10:00 AM – 4:00 PM (Mon–Fri)	dean.academics@nit-fictional.edu
HoD, Artificial Intelligence & Data Science	Dr. Nilesh Rao, Head of Department	AI & Data Science Block, Room 201	10:00 AM – 4:00 PM (Mon–Fri)	hod.cse@nit-fictional.edu

All names, room numbers, and contact details listed above are entirely fictional and created for RAG testing purposes only.

12. Frequently Asked Academic Questions

Q. What is the minimum attendance required for examination eligibility?
A. You need at least 75% attendance in each course, calculated course-wise, to be directly eligible for the End-Semester Examination (Section 2.1).

Q. Can medical grounds be used for attendance condonation?
A. Yes. If your attendance is between 60% and 64.9%, you can apply for medical condonation with a certificate from a Registered Medical Practitioner, which relaxes your eligibility threshold to 60% (Section 2.4).

Q. What documents are required for medical condonation?
A. A medical certificate on official letterhead stating diagnosis and dates, and, if you were hospitalized for more than 3 days, certified hospital admission and discharge records (Section 2.4).

Q. How much is the attendance condonation fee?
A. The ordinary condonation processing fee is Rs. 500 per course, payable at the time of application (Section 2.3).

Q. What happens if my attendance is below the permitted limit even after condonation?
A. You are marked “Detained” in that course, cannot sit the End-Semester Examination, and must re-register and repeat the course in a later offering (Section 2.6).

Q. What is the passing grade?
A. Grade “D”, corresponding to at least 40 marks out of 100 and a grade point of 4, is the minimum passing grade in every course (Section 4.2).

Q. How is CGPA calculated?
A. CGPA is the credit-weighted average of your SGPA across all completed semesters: $CGPA = \Sigma(SGPA_k \times TC_k) \div \Sigma TC_k$ (Section 4.3).

Q. What happens if I fail a Semester III course?
A. You receive grade “F” and can clear the course through the Backlog Examination in the next regular offering, provided your original attendance met the Section 2 requirement (Section 3.7).

Q. What is the late assignment submission penalty?
A. A flat 5% deduction per working day of delay applies, up to a maximum of five working days, after which the assignment scores zero (Section 8.2).

Q. What topics are covered in ADA501?
A. ADA501 (Big Data Systems and Architectures, 4 credits) covers distributed storage, batch processing, stream processing, NoSQL/columnar stores, and cluster resource management (Section 6.1).

Q. When is the Semester III End-Semester Examination?
A. The Semester III End-Semester Examination runs from November 24 to December 5, 2025 (Section 9.1).

Q. Is there a supplementary examination if I miss the End-Semester Examination on medical grounds?
A. Yes, a Supplementary Examination is ordinarily conducted within 6 weeks of the original End-Semester Examination for eligible medical-absence cases (Section 3.6).

Q. Can I get an extension on an assignment deadline?
A. You may request up to a 3 working-day extension in writing at least 24 hours before the deadline; medically supported extensions may be longer at the instructor's discretion (Section 8.3).

Q. What is the deadline to apply for grade revaluation?

A. Revaluation requests must be submitted within 10 working days of result declaration, along with the prescribed fee of Rs. 600 per script (Section 3.7 and Section 10.7).

Q. Who approves an attendance condonation application?
A. The Dean of Academic Affairs approves condonation applications on the written recommendation of the Head of Department (Section 2.3).

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